

SUNNY ZHANG

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EDUCATION

University of Toronto, St. George

Honours BSc, Double Major in Computer Science and Mathematics; GPA: 3.7/4.0

Toronto, ON

2025 – 2029

EXPERIENCE

AI Research Intern

Jun 2026 – Sep 2026

Algoverse AI Research Program

Remote

- Architected a 4-agent Llama-3.3-70B hierarchical trading-firm simulator: 120 episodes, 2,400 logged decisions.
- Defined the methodology, ablation grid, and harness the team built against, and ran most experiments, separating transcripts from action traces so each scored independently.
- Lead author of a 3-person team, resulting in the preprint: managers concealed competitive pressure in language while acting on it, so transcript-only monitoring missed the behavior entirely.

PROJECTS

LaunchPilot — Hack Canada 2x Track Winner | *Next.js, TypeScript, FastAPI, Postgres, Docker*

- Built an AI launch assistant in 36 hours: 4 agents research a market, write positioning, generate assets, and queue outreach email behind a human approval gate.
- Owned agent orchestration and the inter-agent communication pipeline, threading shared project context through persistent stage threads so each agent builds on the last instead of restarting cold.
- Owned the auth system end to end: Auth0 JWT validation, scope-gated API routes, and the frontend session flow, plus a dev mode keeping the 49-endpoint API runnable without an Auth0 tenant.

Tiny Society | *Next.js, TypeScript, FastAPI, Supabase, Docker*

- Built a world-grounded multi-agent simulator that turns any user-authored setting into a 7-day narrative through planning, memory retrieval, reflection, and relationship consequences.
- Architected a single-pass World Lens to extract graph facts, affordances, gathering places, and narrative register once, then ground every agent prompt and world-fit character proposal.
- Designed a deterministic 3-tier witness model to replace platform-specific action channels while preserving bounded LLM cost, legacy-save compatibility, and byte-identical batch and incremental runs.

Feedback or Annealing? | *Python, Numba, pytest, mypy*

- Benchmarked 3 aggregation rules on a 9,261-state MDP over 20 paired seeds and a matched control.
- Hardened Numba kernels with a 96-test suite run compiled, in pure Python, and bounds-checked on a cold cache, catching out-of-range reads a warm cache silently hides.
- Preregistered negative result: feedback gains at most 0.2%, and plain value iteration converges in 4.20M backups against 9.2M+ for every aggregation run.

Parallel State Aggregation | *Python, NumPy, Numba, Matplotlib*

- Built a multithreaded value-iteration benchmark in Numba over 40,000–1,000,000-state MDPs, 1–10 threads, and paired seeds.
- Traced the thread-scaling ceiling to memory bandwidth and a serial rebinning step, and exposed 99.67% of the work the original instrumentation was not counting.
- Measured plain value iteration hitting the error target $1.3\text{--}2.2\times$ faster than aggregation across all 5 configurations.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Java, C++

Web: FastAPI, Flask, Next.js, React, PostgreSQL, SQLAlchemy, Alembic

Testing & Infrastructure: pytest, GitHub Actions, Docker, Git, Fly.io, Vercel, Supabase

ML & Data: PyTorch, NumPy, Pandas, Numba